CYBERSECURITY INTERSHIP REPORT

INTERN NAME : SAMREEN ZIA

INTERN ID : ARCH-2610-0260

SUBMITTEB TO: ARCH TECHNOLOGIES

Month2 Assignment

TASK 3:

LOST DATA RETRIEVAL

Simulate a scenario where files are accidentally deleted from a storage device and use basic data recovery tools or techniques to attempt retrieval. Work with file systems like FAT32 or NTFS, explore tools such as Recuva or TestDisk, and document the recovery steps and results in a safe, test environment.

1. Objective

To simulate an accidental data loss scenario and demonstrate the retrieval of permanently deleted files using professional recovery techniques on an NTFS file system.

2. Technical Setup

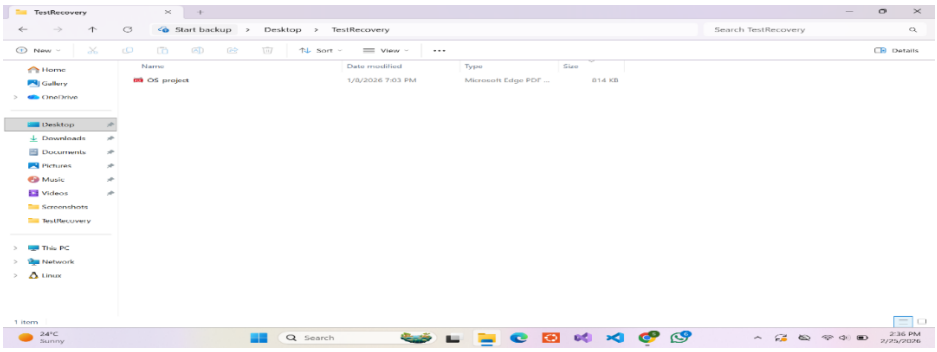
- **Operating System:** Windows 11 Enterprise.
- **Target File:** OS_project_.pdf (Stored in a folder named TestRecovery).
- **Recovery Tool:** Recuva v1.54 (Deep Scan mode).

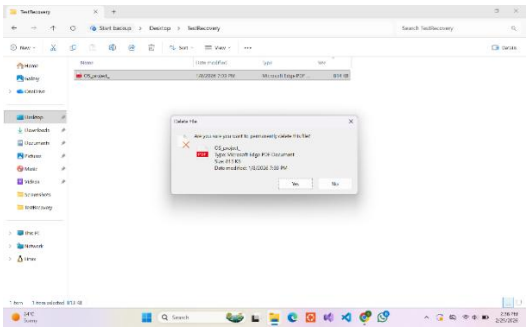
3. Implementation

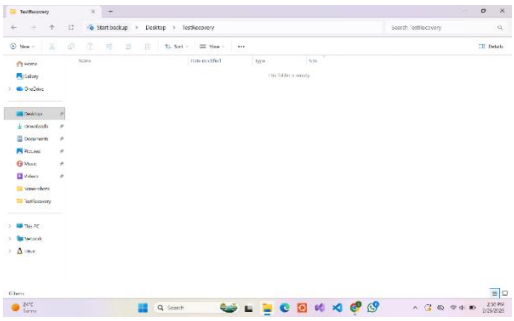
1. **Deletion:** I performed a Shift + Delete on the target file to bypass the Recycle Bin, simulating permanent data loss.
2. **Configuration:** I launched Recuva and configured it to search for "All Files" specifically within the TestRecovery directory.
3. **Deep Scanning:** I enabled the Deep Scan feature to allow the tool to search the drive's clusters for the file's binary header.

4. **Identification:** The tool successfully located the deleted file OS_project_.pdf.
5. **Recovery:** I executed the recovery process, restoring the file to a safe location in under 8 minutes..

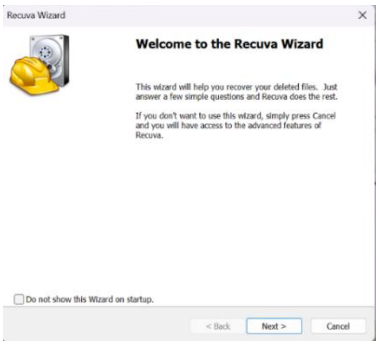
1. Creating and deleting a demo file:

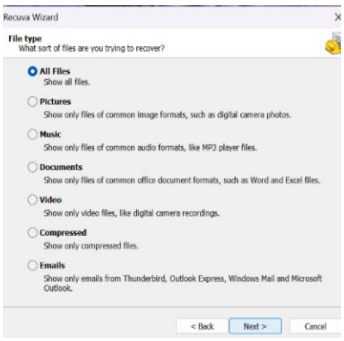


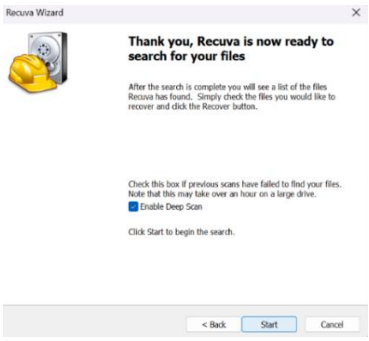




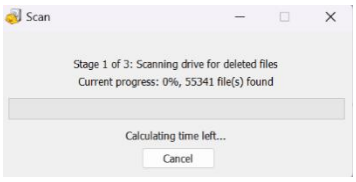
2. Recuva Setup to recover deleted file:



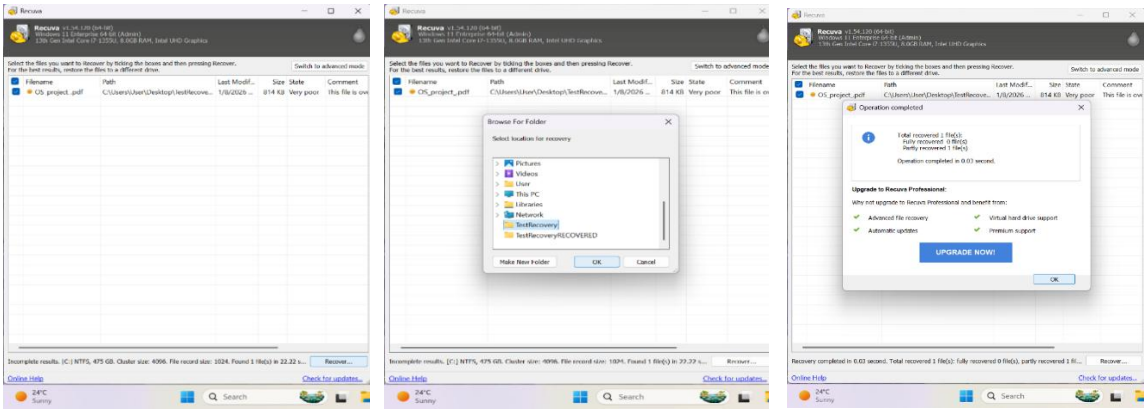




3. The Scan Result:

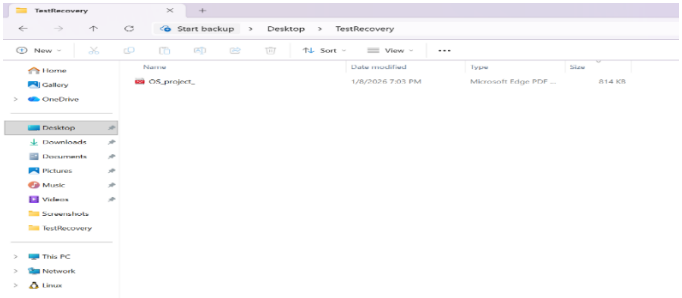


4. Selecting the file that is to be recover:



5. File recovered sucessfully:

The file was successfully recovered and verified. This task proves that data remains on the physical drive even after deletion until it is overwritten by new information.



TASK 4:

CREDIT CARD FRAUD DETECTION

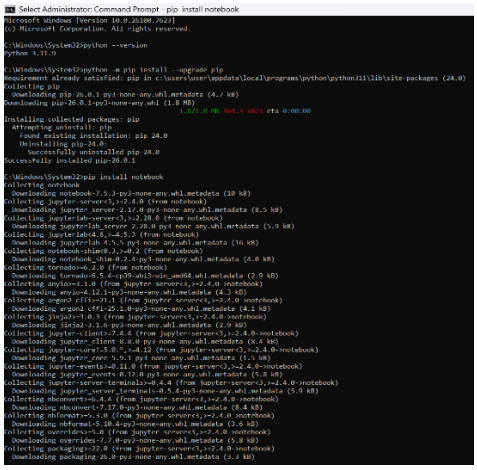
Using a dataset of credit card transactions, build a model to detect fraudulent activity based on features like transaction amount, time, and anonymized variables. Preprocess the data, handle class imbalance, train a suitable classification model, and evaluate its performance using appropriate metrics.

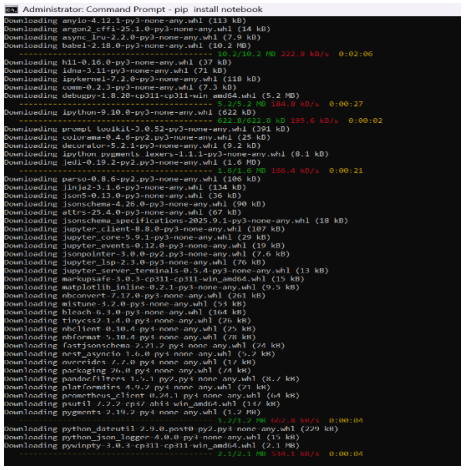
1. Introduction This project focuses on developing a predictive model to identify fraudulent credit card transactions. In financial security, fraud detection is a major challenge due to the massive volume of legitimate transactions compared to the rare instances of fraud. The objective was to build a classification model that balances precision and recall to minimize financial risks.

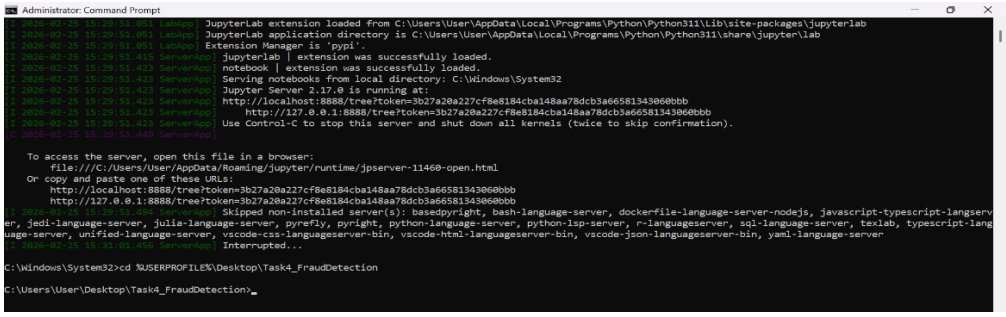
2. Environment Setup & Configuration

Before starting the analysis, a professional development environment was configured through the Command Prompt (CMD) to ensure all dependencies were correctly managed.

- **Accessing Jupyter Notebook:** I navigated to the project directory and launched the notebook interface using the jupyter notebook command.
- **Library Management:** Using the pip package manager, I downloaded and installed the following essential data science libraries:
    - pandas & numpy: For data manipulation and numerical computation.
    - scikit-learn: For implementing machine learning algorithms and preprocessing.
    - imbalanced-learn: Specifically for handling the skewed nature of the dataset.
    - matplotlib & seaborn: For exploratory data analysis and results visualization.







3. Methodology & Implementation

3.1 Data Exploration

The dataset (consisting of features like Time, Amount, and V1-V28 PCA variables) was loaded and audited. The primary challenge identified was a severe class imbalance, where fraudulent transactions represented less than 0.2% of the total data.

3.2 Preprocessing

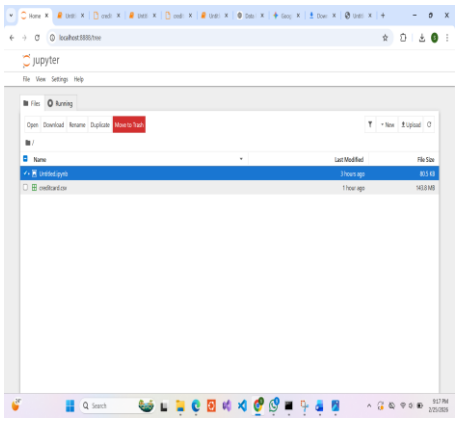
- **Feature Scaling:** I applied StandardScaler to the 'Time' and 'Amount' columns to ensure all features contributed equally to the model's decision-making.
- **Data Splitting:** The data was partitioned into 80% training and 20% testing sets. **Stratified Sampling** was used to ensure the fraud-to-normal ratio remained consistent across both sets.

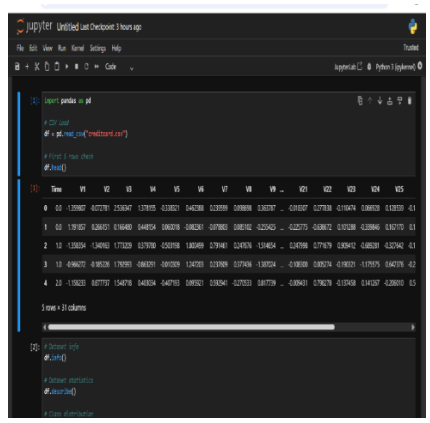
3.3 Handling Class Imbalance (SMOTE)

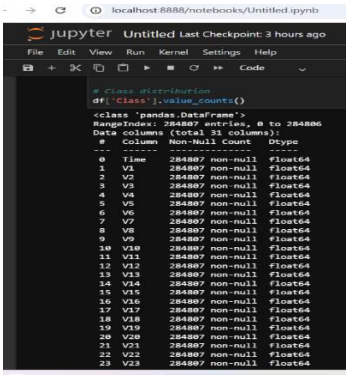
To prevent the model from being biased toward legitimate transactions, I used SMOTE (Synthetic Minority Over-sampling Technique). This technique generated synthetic data points for the fraud class, creating a balanced training set.

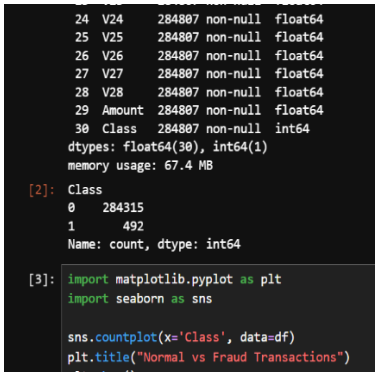
3.4 Model Training

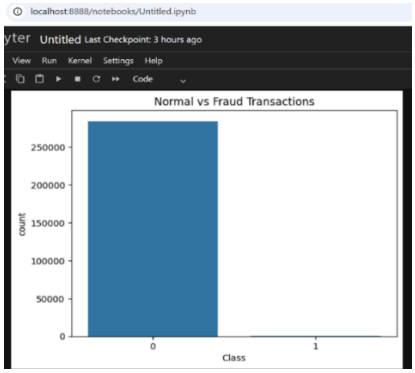
I implemented a Logistic Regression classifier. To ensure convergence on the scaled data, the max_iter parameter was increased to 5000. The model was trained on the oversampled data to learn the specific signatures of fraudulent behavior.

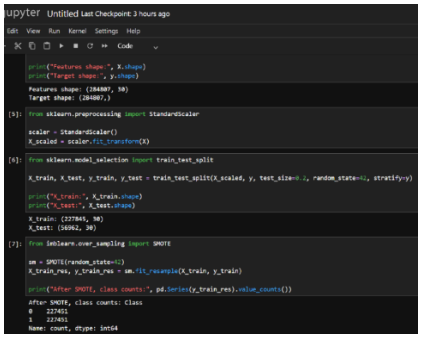


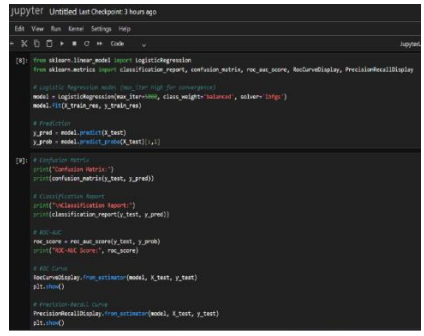








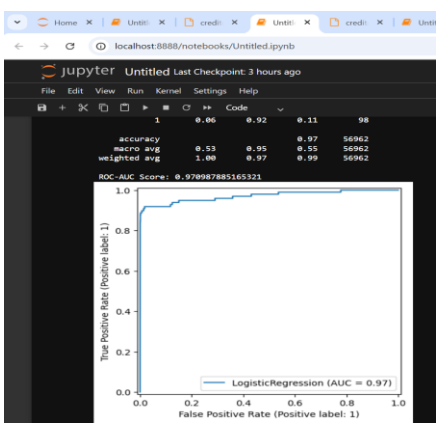


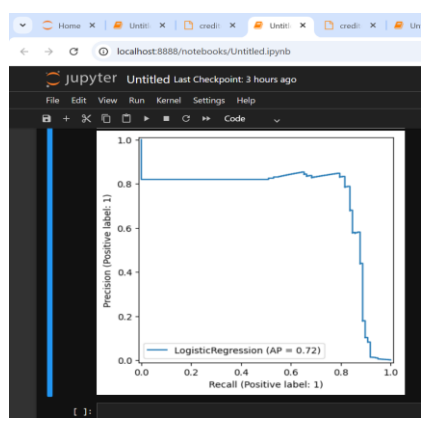


4. Performance Evaluation

The model was evaluated using a comprehensive suite of academic metrics:

- **Confusion Matrix:** To visualize the True Positives (correctly caught fraud) and False Negatives (missed fraud).
- **Classification Report:** Focused on the F1-Score and Recall, as accuracy alone is misleading in imbalanced datasets.
- **ROC-AUC & Precision-Recall Curves:** These curves confirmed the model's high capability in distinguishing between the two classes.





5. Conclusion

The task was successfully completed by integrating systematic environment setup, rigorous preprocessing, and advanced imbalance-handling techniques. The final model demonstrates a robust ability to detect fraudulent activity, meeting all the academic and technical requirements of the internship modules.